复旦数学资料汇编
动力系统

来源：香蕉空间 Solution 资料库

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第一章　动力系统

1.1　试卷: 动力系统

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Autumn 2025 Exam: Dynamical systems (李嘉伦)

You can use all the results from the course and please write clearly the name or the content of the result. Each sub-question has 10 points.

Exercise 1 (Unique ergodicity) • Give a definition of unique ergodicity of a continuous dynamical system $f : X \to X$ on a compact metric space.

- Give one example of uniquely ergodic system.
- If (f, X) is a uniquely ergodic system, show that for any continuous function φ, the $S_n\varphi(x)$ converges uniformly for $x \in X$.

Exercise 2 (Symbolic dynamics) Let $\Lambda = \{1, 2, \cdots, k\}$ be a finite set, $\Omega = \Lambda^{\times\mathbb{N}}$ and S is the shift map on Ω given by $S(x_n)_n = (x_{n+1})_n$.

- What are the periodic points of S?
- Let P_n be the set of periodic points of period n (that is $S^n x = x$), what is $\#P_n$, the cardinality of P_n?
- What is the weak limit of the following measures?

$$\lim_{n\to\infty} \frac{1}{\#P_n} \sum_{x\in P_n} \delta_x$$

- If we let P'_n be the set of primitive periodic points with period n (that is if $S^m x = x$ then n divides m), show the same convergence as the previous one when replacing P_n by P'_n.

Exercise 3 (Entropy) We write T_p for the times p map on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Let μ be a T_2 invariant measure on $\mathbb{T}$.

- Show

$$h_{(T_3)_*\mu}(T_2) \le h_\mu(T_2)$$

- Show

$$h_\mu(T_2) = h_{(T_3)_*\mu}(T_2)$$

Exercise 4 (Torus and Subtorus) Let $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, and consider the map f_M on the torus $\mathbb{T}^3$

given by

$$f_M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = M \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mod \mathbb{Z}^3.$$

- Is the Lebesgue measure on $\mathbb{T}^3$ ergodic for f_M?
- For any $t \in \mathbb{T}$, show the measure $d\mu_t = dxdy$ on the subtorus $z = t$ is ergodic for f_M.
- Show it is also mixing for f_M.

What about the same questions for $M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$?